Ratio, Root, and Alternating Series Tests

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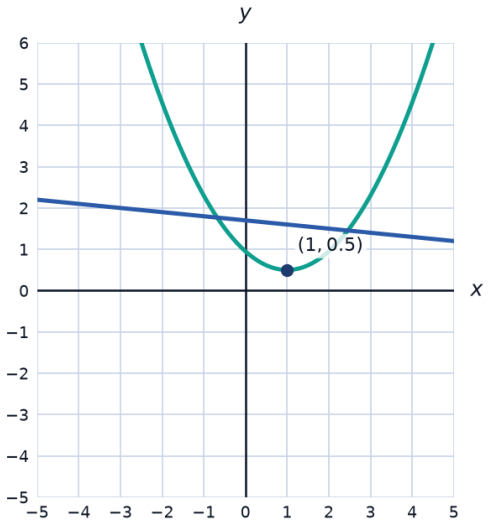
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: The Ratio Test**

Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.

2 **[Foundational] • Target: The Root Test**

Use the Root Test to determine whether $\sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$ converges.



3 **[Foundational] • Target: Alternating series and the Alternating Series Test**

State the Alternating Series Test, and use it to show $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges.

4 **[Foundational] • Target: Alternating series and the Alternating Series Test**

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n+1}$ converges using the Alternating Series Test.

5 **[Foundational] • Target: Alternating series error bound**

Approximate $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ using the first 4 terms, and bound the error using the Alternating Series Error Bound.

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6 [Foundational] • Target: Absolute vs. conditional convergence

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ converges absolutely, conditionally, or diverges.



7 [Foundational] • Target: More Ratio Test practice

Explain why the Ratio Test is INCONCLUSIVE for $\sum_{n=1}^{\infty} \frac{1}{n}$, and identify which test should be used instead.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Partial sums of an alternating series (visualizing oscillating convergence)

Compute the first 5 partial sums of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ (the alternating harmonic series, which converges to $\ln 2 \approx 0.693$), and describe how they oscillate.

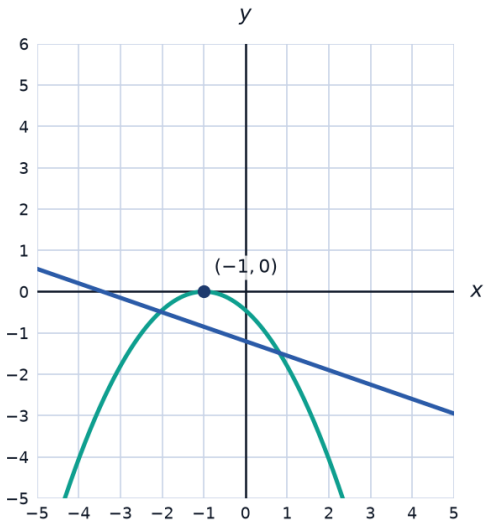
9 [Intermediate] • Target: Determining the number of terms needed for a given accuracy

For $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3}$, find the minimum number of terms needed so the partial sum is within 0.001 of the true sum, using the Alternating Series Error Bound.

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10 **[Intermediate]** • **Target: Mixed practice: classify each series**

Classify each series as convergent (state absolutely or conditionally) or divergent:



a $\sum \frac{(-1)^n}{\sqrt{n}}$

b $\sum \frac{(-1)^n n^2}{n^3+1}$

c $\sum \frac{(-1)^n}{n!}$

11 **[Intermediate]** • **Target: Why the Ratio Test fails for p-series**

Show that the Ratio Test is inconclusive for EVERY p-series $\sum \frac{1}{n^p}$, regardless of p.

12 **[Intermediate]** • **Target: Combining multiple tests in one problem**

Level 3 · Advanced / AP FRQLet $f(x)$ be a function whose third derivative satisfies $|f'''(x)| \le 6$ for all $x \in [0, 0.4]$. If the second-degree Taylor polynomial $P_2(x)$ centered at $c = 0$ is used to approximate $f(0.2)$, find the maximum possible error in this approximation using the Lagrange Error Bound.

13 **[Intermediate]** • **Target: The Ratio Test**

[Beyond AB — Core BC Topics Overview · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

14 **[Intermediate]** • **Target: The Root Test**

[Unit 9 · Parametric Equations & Vector-Valued Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

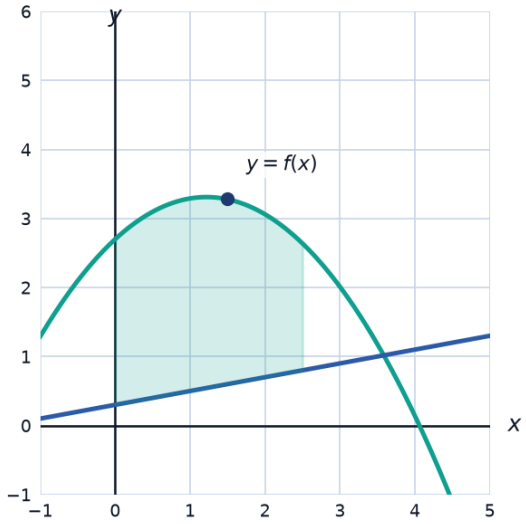
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15 [Challenging] • Target: Alternating series and the Alternating Series Test • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Level 2 · MediumFind the minimum number of subintervals n required to guarantee that the Trapezoidal approximation of $\int_1^3 \frac{1}{x}\,dx$ has an absolute error less than 10^{-4}.

16 [Challenging] • Target: Alternating series error bound • Cross-Lesson: 3.01 Parametric Equations

Level 3 · Advanced / AP FRQEvaluate the improper integral $\int_0^\infty xe^{-2x}\,dx$ or establish that it diverges.



17 [Challenging] • Target: Absolute vs. conditional convergence • Cross-Lesson: 3.02 Vector-Valued Functions

Does $\sum_{n=0}^{\infty} (0.9)^n$ converge, and to what?

18 [Challenging] • Target: More Ratio Test practice • Cross-Lesson: 4.01 Polar Curves and Derivatives

Level 1 · FoundationsEvaluate the indefinite integral $\int \arctan(x)\,dx$.

19 [Challenging] • Target: Partial sums of an alternating series (visualizing oscillating convergence) • Cross-Lesson: 4.02 Area in Polar Coordinates

Level 1 · FoundationsFor the parametric curve $x(t) = \cos(t)$ and $y(t) = \sin(2t)$, find all values of $t \in [0, \pi]$ where the curve has a vertical tangent line.

20 [Challenging] • Target: Determining the number of terms needed for a given accuracy • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Level 2 · MediumA particle moves with position vector $\vec{r}(t) = \langle t^2 - 4t, \frac{1}{3}t^3 - 4t\rangle$. At what time $t > 0$ is the particle at rest?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The Ratio Test

Use the Ratio Test to find $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for $\sum_{n=1}^{\infty} \frac{n}{3^n}$.

a. 3

b. $\frac{1}{3}$

c. $\frac{4}{3}$

d. $\frac{1}{4}$

22 [Foundational] • Target: The Root Test

Use the Root Test to find $L = \lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{4n+4}{11n-7}\right)^n\right|}$.

a. $\frac{4}{11}$

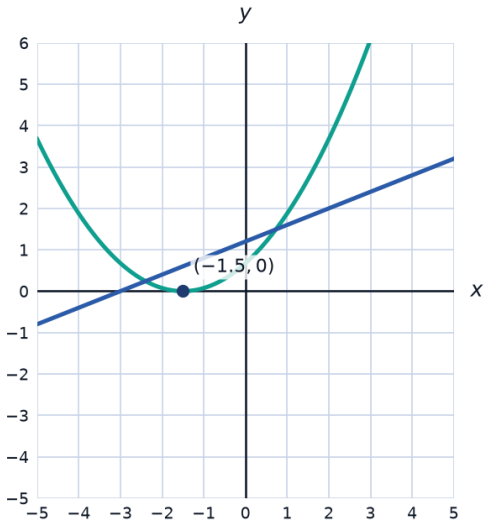
b. $\frac{11}{4}$

c. $\frac{19}{22}$

d. $\frac{15}{11}$

23 [Foundational] • Target: Alternating series and the Alternating Series Test

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ needed to guarantee an error less than $\frac{1}{10000}$?



a. 8

b. 9

c. 7

d. 10000

24 [Foundational] • Target: Alternating series error bound

When approximating $\sin(0.3)$ by its Taylor polynomial of degree 5 about $x = 0$, what is the alternating series error bound?

a. $\frac{(0.3)^7}{(7)!}$

b. $\frac{1}{(7)!}$

c. $\frac{(0.3)^5}{(5)!}$

d. $\frac{(0.3)^7}{(3)!}$

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25 [Foundational] • Target: Absolute vs. conditional convergence

The ratio test gives $L = 1$: the test is:

a. inconclusive

b. divergent

c. convergent

d. absolutely convergent

26 [Foundational] • Target: More Ratio Test practice

Error bound: $|\cos x - 1 + x^2/2| \le x^4/24$ (from T_4 bound). This means the error at $x = 0.1$ is at most:

a. $0.0001/24$

b. $0.1^4/24$

c. $0.1^2/2$

d. $1/24$

27 [Foundational] • Target: Partial sums of an alternating series (visualizing oscillating convergence)

Series $\sum_{n=1}^{\infty} (-1)^{n-1}/\sqrt{n}$ converges by alternating series test. Is it absolutely convergent?

a. Yes, $p = 1/2 < 1$ diverges

b. No, $\sum 1/\sqrt{n}$ diverges ($p = 1/2 \le 1$)

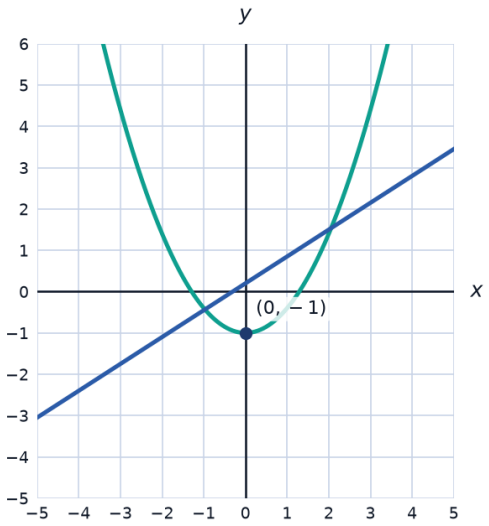
c. Yes, $\sum 1/\sqrt{n}$ converges

d. Cannot determine

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28 [Foundational] • Target: Determining the number of terms needed for a given accuracy

Which test determines $\sum n^2/e^n$? Answer: by ratio test $L =$



a. e

b. 1

c. $1/e < 1$

d. e^2

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Mixed practice: classify each series

Series $\sum_{n=1}^{\infty} n \cdot (1/2)^n$: ratio test gives $L =$

a. 1

b. 2

c. $1/2$

d. $n/2$

30 [Intermediate] • Target: The Root Test with more complex expressions

Alternating test: $a_n = 1/n!$: does $a_n \to 0$? Does $a_{n+1} < a_n$?

a. Neither

b. Both $a_n \to 0$ and $a_{n+1} < a_n$

c. $a_n \to 0$ only

d. $a_{n+1} > a_n$

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31 [Intermediate] • Target: Why the Ratio Test fails for p-series

The integral $\int_0^1 e^{-x^2}dx$ (error function) — the exact value cannot be expressed in elementary functions. Using the series, approximate as $\sum \frac{(-1)^n}{n!\,(2n+1)}$. First two terms:

a. $1 - 1/6 \approx 0.833$

b. $1 - 1/3 \approx 0.667$

c. $1 + 1/3 \approx 1.333$

d. 0

32 [Intermediate] • Target: Combining multiple tests in one problem

Maclaurin series for $\tan^{-1} x$ at $x = 1$ gives $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \cdots$. This converges very slowly. The 10th partial sum approximates $\pi/4$ with error less than:

a. $1/11$

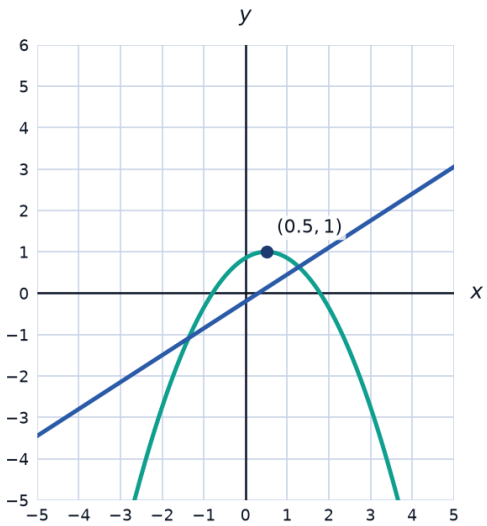
b. $1/21$

c. $1/19$

d. 0.01

33 [Intermediate] • Target: The Ratio Test

Absolute convergence implies conditional convergence. The converse:



a. is also true

b. is false (counterexample: alternating harmonic series)

c. requires $a_n \geq 0$

d. is true for power series

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34 **[Intermediate]** • **Target: The Root Test**

The Maclaurin series for a function $f(x)$ is given by $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$. If the first three terms of the Maclaurin series for $f(x)$ are $5 - 2x + \frac{3}{2}x^2$, what are the values of $f(0)$, $f'(0)$, and $f''(0)$?

a. $f(0) = 5, f'(0) = -2, f''(0) = 3$

b. $f(0) = 5, f'(0) = -2, f''(0) = 6$

c. $f(0) = 5, f'(0) = -1, f''(0) = 3$

d. $f(0) = 3, f'(0) = -2, f''(0) = 5$

35 **[Intermediate]** • **Target: Alternating series and the Alternating Series Test**

Consider the Taylor series expansion of the function $f(x) = \frac{1}{1-x}$ centered at $x = 0$. If the sum of the first three non-constant terms of the Taylor series is denoted by $S_3(x)$, and we are given that $S_3(x) = ax + bx^2 + cx^3$, find the value of $b + c$.

a. 1

b. 2

c. 3

d. 4

36 **[Intermediate]** • **Target: Alternating series error bound**

The error function $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}dt$. Using $e^{-t^2} = \sum(-1)^n t^{2n}/n!$:

a. $\operatorname{erf}(x) = \sum x^n/n!$

b. $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}}\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{n!\,(2n+1)}$

c. $\operatorname{erf}(x) = e^{-x^2}$

d. $\operatorname{erf}(x) = 2x/\sqrt{\pi}$

37 **[Intermediate]** • **Target: Absolute vs. conditional convergence**

Generalised binomial: $\sqrt{1-x} = \sum_{n=0}^{\infty}\binom{1/2}{n}(-1)^n x^n$. First three terms:

a. $1 - x - x^2$

b. $1 + x/2 - x^2/8$

c. $1 - x/2 - x^2/8$

d. $1 - x/2$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: More Ratio Test practice • Cross-Lesson: 4.01 Polar Curves and Derivatives

Use alternating series estimation: $|\cos 1 - 1 + 1/2| \le |-1/4!\,| =$

a. 1/12

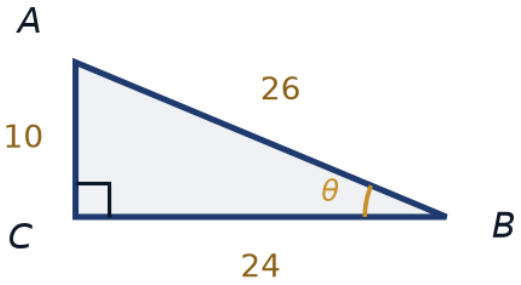
b. 1/120

c. 1/6

d. 1/24

39 [Challenging] • Target: Partial sums of an alternating series (visualizing oscillating convergence) • Cross-Lesson: 4.02 Area in Polar Coordinates

Approximate $\int_0^1 \frac{\sin x}{x} dx$ using power series (first three terms):



a. $1 - 1/3 + 1/10$

b. $1 - 1/18 + 1/600$

c. $1 - 1/6 + 1/20$

d. $1 - 1/18$

40 [Challenging] • Target: Determining the number of terms needed for a given accuracy • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Convergence of $\sum n^2/3^n$: by ratio test, $L =$

a. 1

b. 1/3

c. 3

d. 0

41 [Challenging] • Target: Mixed practice: classify each series • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Integral test convergence of $\sum_{n=1}^{\infty} 1/n(\ln n)^2$ (from $n = 2$) gives sum approximately:

a. converges

b. diverges

c. equals e

d. equals π^2

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42 [Challenging] • Target: The Root Test with more complex expressions • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Error bound for alternating series: $|S - S_n| \leq$

a. a_n

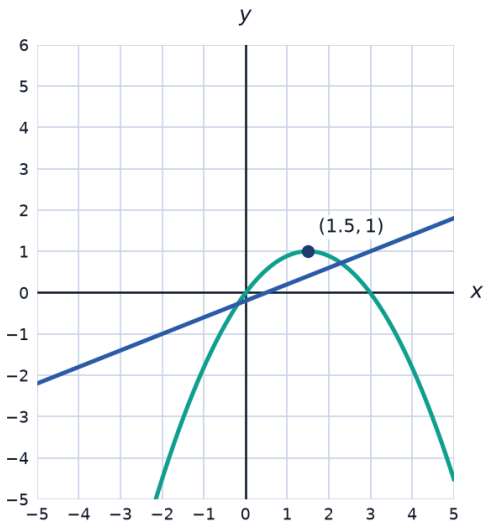
b. S_n

c. na_n

d. a_{n+1}

43 [Challenging] • Target: Why the Ratio Test fails for p-series • Cross-Lesson: 1.01 Integration by Parts

Dirichlet test: $\sum a_n b_n$ converges if a_n monotone $\to 0$ and partial sums of $\sum b_n$ are:



a. convergent

b. zero

c. increasing

d. bounded

44 [Challenging] • Target: Combining multiple tests in one problem • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

The error in using $T_n(x)$ decreases as n increases (assuming convergence). This is because the remainder $R_n \to$

a. 1

b. $f(x)$

c. $n!$

d. 0

45 [Challenging] • Target: The Ratio Test • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Use the integral test to show $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ converges. The integral $\int_2^{\infty} \frac{dx}{x(\ln x)^2} =$

a. ∞

b. π

c. $\ln 2$

d. $1/\ln 2$

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